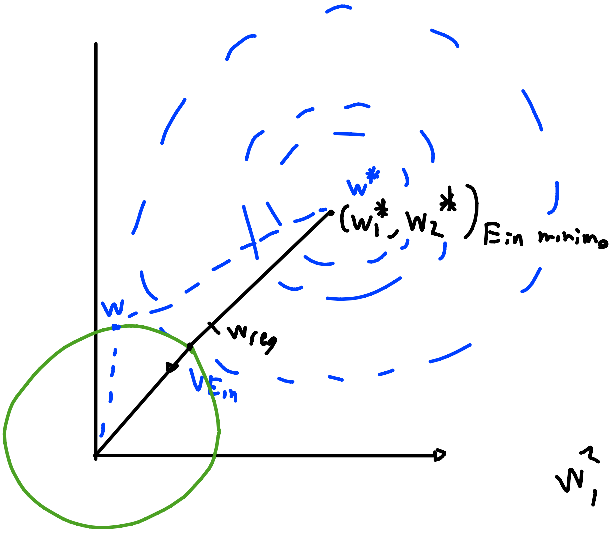

$w^*, b^* = \arg\min_{w,b} E_{in}(w, b)$

baj 0

$\sum_{j=1}^{n} w_j^2 \le C$

$w^T w \le C$

$\|w\|_2^2 \le C$

$w_1^2 + W_2^2 \le \sqrt{C}^2$

$w_1^2 + w_2^2 = C$

$w_{reg} = -k \nabla_w E_{in}(w, b)$

$2\lambda W_{reg} + \nabla E_{in}(w_{reg}, b) = 0$

$\nabla_w (E_{in}(w, b) + \lambda w^T w)$

$\nabla E_{in}(w, b) + 2\lambda w$